UTP Vendor Code Requisition Documentation

Project Name : UTP Vendor Code Requisition (UTP VCR)
Version : 1.0.0
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1. System Overview

The **UTP Vendor Code Requisition** is a centralized web-based platform designed to streamline the vendor registration and code requisition process at Universiti Teknologi PETRONAS. It replaces manual, fragmented workflows with a secure, digital ecosystem that manages document submission, identity verification, and approval workflows.

Key Objectives

- **Confidentiality:** Secure handling of sensitive user data (IC, Bank Statements) via segregated access and encryption.
- **Efficiency:** Automated checks (e.g., 3-month bank statement validity) to reduce administrative workload.
- **Centralization:** A single source of truth for Student, External, and Company vendor code requisition.

2. User Roles & Scope

A. End Users (Applicants)

- **Student:** FYP/Postgrad students requiring vendor codes for claims. Submits FYP Form 02.
- **External Appointment (EA):** Individual consultants/experts. Must select a referring AE Admin.
- **Company:** Corporate entities. (Phase 2 Implementation).

B. Administrators

- **KYC6 Admin:** Compliance officer. Verifies Identity (IC vs Face).
- **AE Admin:** Internal UTP staff members who serve as the point of reference for external applicants.
- **Master Admin:** Super-user. Has final authority to generate Vendor Codes, verify documents, and archive records.

3. Technical Architecture

Server Requirements

- **Web Server:** Apache (via XAMPP/Laragon) or Nginx.
- **Language:** PHP 8.0+ (Native/Plain PHP).
- **Database:** MySQL / MariaDB.
- **Dependencies:**
    - Composer (Dependency Manager)
    - PHPMailer (For email notifications)
    - vlucas/phpdotenv (For environment variable management)

Folder Structure

```
UTP_Vendor_Hub/
├── Admin/                # Administrative Portal (Secured)
│   ├── Master_Admin/     # Super Admin logic & Zip generation
│   ├── KYC6_Admin/       # Compliance verification logic
│   ├── AE/               # AE logic
│   └── ...
├── user/                 # Public/Vendor Portal
│   ├── signUp.php        # Registration logic
│   ├── user_homePage.php # Dashboard
│   └── ...
├── config.php            # Database connection settings
├── vendor/               # Composer dependencies (PHPMailer, DotEnv)
├── generated_zips/       # Temp storage for admin downloads
└── uploads/              # Secure storage for user documents
```

4. Installation & Setup Guide

Step 1: Database Setup

1. Access phpMyAdmin or your MySQL interface.
2. Create a new database named **UTP_Vendor_Hub.**
3. Import the provided SQL file: **database/UTP_Vendor_Hub.sql.**

Step 2: Configuration

1. Open **config.php.**
2. Update the database credentials if necessary:

```
$db_host = 'localhost';
$db_user = 'root'; // Change for production
$db_pass = “ ”;     // Change for production
$db_name = 'UTP_Vendor_Hub';
```

3. Create a **.env** file in the root directory for SMTP settings:

```
SMTP_USER=your_email@gmail.com
SMTP_PASS=your_app_password
```

Step 3: Deployment

1. Upload all files to the server's **public_html** directory.
2. Ensure the **uploads/** and **generated_zips/** directories have **write permissions (755 or 777)**.
3. Verify that **composer install** has been run to generate the **vendor/** folder.

5. Security Features

- **Segregated Portals:** Admin and User logic are kept in strictly separate directories to minimize unauthorized access.
- **Input Sanitization:** All database inputs are sanitized using mysqli_real_escape_string and Prepared Statements to prevent SQL Injection.
- **Password Hashing:** User passwords are encrypted using password_hash() (Bcrypt).
- **Session Management:** Strict session checks prevent unauthorized URL access and enforce role-based access control (RBAC).

6. Constraints & Known Limitations

- **Email Delivery:** Due to UTP's strict SMTP firewall policies, the system is currently configured to use an external SMTP service (Gmail) for notifications. Internal @utp.edu.my relay requires IT whitelisting.
- **Company Data:** Automated SSM verification is not implemented due to API licensing costs.

7. Future Roadmap

- Integration with UTP's official LDAP/Active Directory for Single Sign-On (SSO).
- API integration with SSM for real-time company verification.
- Migration to a dedicated PHP Framework (Laravel) for long-term scalability if user load exceeds projected limits.

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